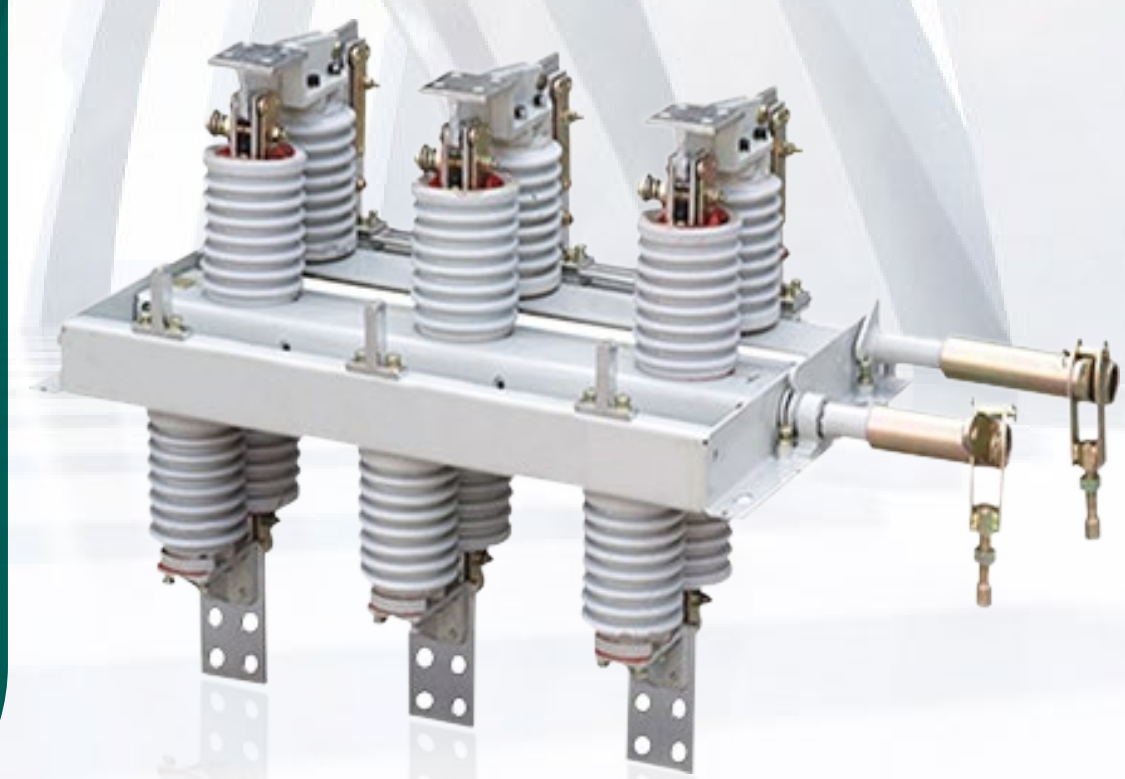
Installation and Operation Manual

GreenPower

GN30/GN19 Series

12/24kV Indoor Disconnectors





About Us»

GreenPower is a manufacturer of medium- and low-voltage switchgear and components, with in-house engineering, manufacturing and testing capabilities. Our manufacturing entity, Zhejiang GreenPower Electric Co., Ltd., was established in 1992. International business and export operations are handled by Zhejiang GreenPower I&E Co., Ltd.

Our product portfolio covers medium-voltage switchgear from 12 kV to 40.5 kV, including air-insulated metal-clad switchgear (AIS), cubicle-type gas-insulated switchgear (C-GIS), ring main units (RMU), vacuum circuit breakers, and a broad range of switchgear components and accessories. We also manufacture low-voltage withdrawable switchgear systems with ratings up to 6300 A.

Engineering and manufacturing are central to what we do. Our production and testing capabilities include automated sheet-metal processing, CNC laser cutting, 3D five-axis laser welding for stainless-steel gas tanks, automatic helium leak detection, and a fully shielded partial-discharge test room. These capabilities support consistent manufacturing quality from enclosure fabrication through assembly and final testing.

For customers moving toward lower-impact power distribution systems, GreenPower offers SF6-free solutions including dry-air insulated RMUs and nitrogen-insulated C-GIS, alongside conventional gas-insulated solutions where required. These technologies help customers respond to increasingly stringent environmental requirements for medium-voltage switchgear.

We also support project-specific engineering based on customer drawings and specifications, including non-standard dimensions, special voltage and frequency requirements, high-altitude applications, customized component selection and other application-specific configurations.

Our goal is simple: to be the switchgear manufacturer our customers return to — because the equipment is properly engineered, reliably manufactured and built for long-term service.

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1.1 Overview of GN30-12(D)/GN19-12 disconnectors

The GN30-12 rotary indoor high-voltage disconnector is a new type of rotary contact disconnector. Its main structure is to fix two groups of insulators and contacts on the upper and lower planes of the three-phase common chassis. The switch can be opened and closed by rotating the contact.

The GN30-12D switch is an additional form with grounding knife on the basis of the GN30-12 switch, which can meet the needs of different power systems. This product is compact in design, small in space occupation, strong in insulation capacity, easy to install and adjust, and its performance meets the requirements of GB198589 AC High Voltage Disconnectors and Grounding Switches. It is applicable to indoor systems with rated voltage of 12 kV AC 50Hz and below, and is used for opening and closing circuits with voltage and no load. It can be used together with high-voltage switch cabinet or used separately.

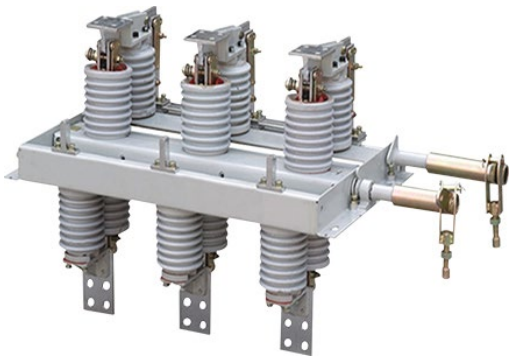
GN19-12 series indoor high-voltage isolation high-voltage switchgear is used in the power system with rated voltage of 12kV, AC 50Hz and below, equipped with GS6-1 type manual operating mechanism for circuit opening and closing under the condition of voltage but no load, and also derivative products such as anti pollution type, plateau type and adding live display device.

1.2 Product model description

GN30-12(D) Indoor rotary high-voltage disconnector

G N 30 - 12 (D) / □ - □

- Thermal stability current (4s)
- Rated current
- With earthing switch
- Rated voltage
- Design number
- Indoor
- Disconnector



GN30-12 series indoor high-voltage disconnectors and GN30-12D type switches are added with earthing blades on the basis of GN30-12 type switches, which can meet the needs of different power systems.

Common models of disconnecting switches in general series are:

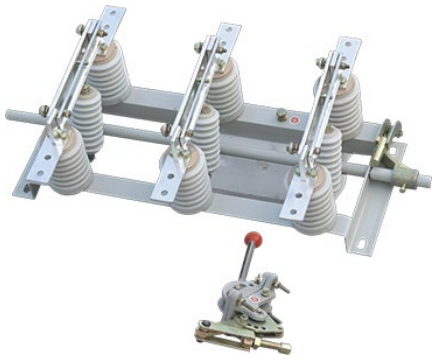
- GN30-12/630A
- GN30-12/1000A
- GN30-12/1250A
- GN30-12/2000A
- GN30-12/3150A
- GN30-12D/630A
- GN30-12D/1000A
- GN30-12D/1250A
- GN30-12D/2000A
- GN30-12D/3150A

In addition, it can also be used as a switching device, as well as an anti pollution type, plateau type and a live display device.

GN19-12 IndoorIndoor rotary high-voltage disconnector

G N 19 - 12 (C) / □ - □

- Rated thermal stability current
- Rated current
- Through wall type
- Rated voltage
- Design number
- Indoor
- Isolating switch



It should be noted that the types of GN19-12 indoor high-voltage disconnectors are divided into GN19-12 flat type; GN19-12C is wall through type (with casing);

Wherein: C1—rotating at the casing side, C2—breaking at the casing side, and C3—both sides are casings. The GN19-12Y flat type is equipped with live display device.

Common models are as follows:

- GN19-12A400A
- GN19-12/630A
- GN19-12/1000A
- GN19-12/1250A
- GN19-12C/400A
- GN19-12C/630A
- GN19-12C/1000A
- GN19-12C/1250A

1.3 Main technical parameters

GN30-12(D) Indoor rotary high-voltage disconnector

Model	Rated voltage (kV)	Maximum operating voltage (kV)	Rated current (A)	Dynamic stability current (kA) (peak value)	4s Thermal stability current (kA) (Valid value)
GN30-12/630A	12	11.5	630	50	20
GN30-12/1000A	12	11.5	1000	80	31.5
GN30-12/1250A	12	11.5	1250	80	31.5
GN30-12/2000A	12	11.5	2000	100	40
GN30-12/2500A	12	11.5	2500	100	40
GN30-12/3150A	12	11.5	3150	100	40
GN30-12D/630A	12	11.5	630	50	20
GN30-12D/1000A	12	11.5	1000	80	31.5
GN30-12D/1250A	12	11.5	1250	80	31.5
GN30-12D/2000A	12	11.5	2000	100	40
GN30-12D/2500A	12	11.5	2500	100	40
GN30-12D/3150A	12	11.5	3150	100	40

GN19-12 Indoor rotary high-voltage disconnector

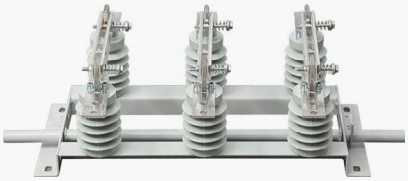
Model	Rated voltage (kV)	Maximum operating voltage (kV)	Rated current (A)	Dynamic stability current (kA) (peak value)	2s Thermal stability current (kA)
GN19-12/400A	12	11.5	400	31.5	12.5
GN19-12/630A	12	11.5	630	50	20
GN19-12/1000A	12	11.5	1000	80	31.5
GN19-12/1250A	12	11.5	1250	100	40
GN19-12C/400A	12	11.5	400	31.5	12.5
GN19-12C/630A	12	11.5	630	50	20
GN19-12C/1000A	12	11.5	1000	80	31.5
GN19-12C/1250A	12	11.5	1250	100	40

1.4 Environmental conditions

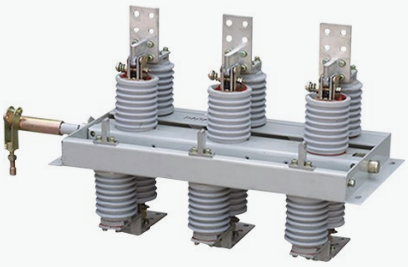
GN30-12(D)Indoor rotary high-voltage disconnector

- The altitude shall not exceed 1000m;
- Ambient air temperature: upper limit+40 ℃ , lower limit - 10 ℃ in general areas; - 25 ℃ in alpine regions;
- Relative air humidity: The daily average value is not more than 95%, and the monthly average value is not more than 90%; The daily average value of saturated steam pressure is not greater than 2.2×10-3MPa, the monthly average value is not more than 1.8×10-3MPa ;
- There is no place with fire, explosion, serious pollution, chemical corrosion and severe vibration.

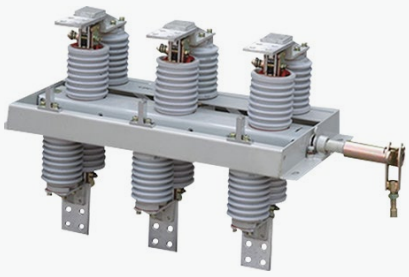
1.5 Photo display of indoor rotary high-voltage disconnector



GN19-12
(Indoor isolating switch)



GN30-12(D)
(Indoor isolating switch)



GN30-12
(Indoor isolating switch)

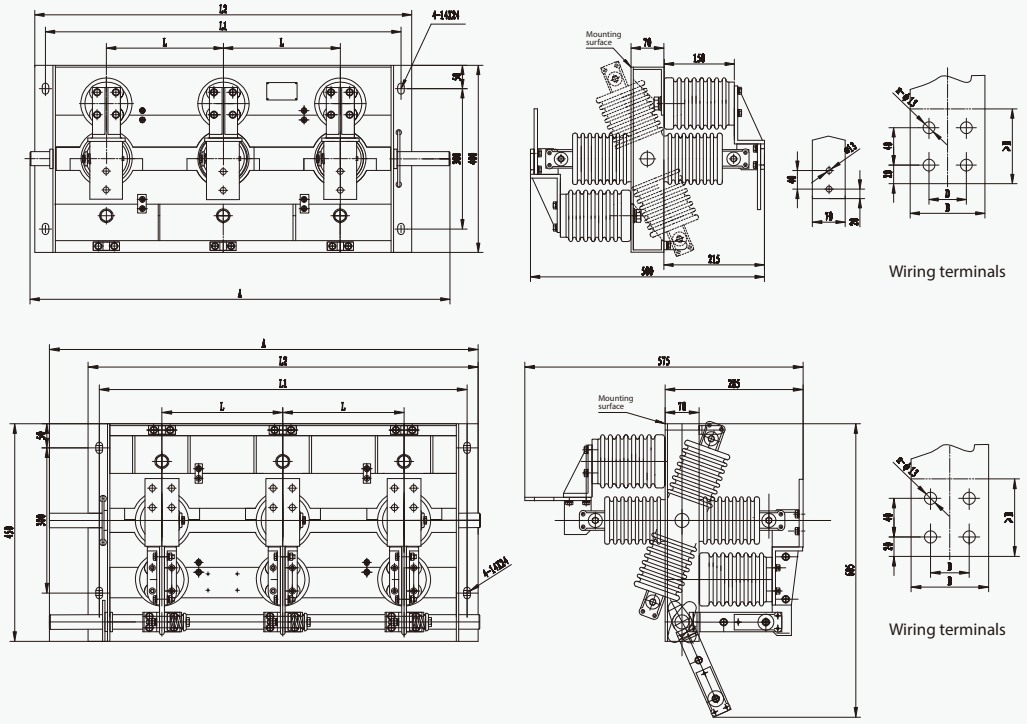
1.6 Product structure and working principle

The conductive parts of each phase of indoor high-voltage disconnectors GN19-12/400A, GN19-12/630A, GN19-12/1000A and GN19-12/1250A are fixed on the base through two post insulators. (For GN19-12C/400A, GN19-12C/630A, GN19-12C/1000A and GN19-12C/1250A, one is a post insulator and the other is a porcelain bushing insulator), three-phase parallel installation is adopted, and the conductive part is composed of contact knife, static contact, contact base (or conducting rod). The middle of each phase contact knife is connected with a pull rod insulator, which is connected with the main shaft installed on the base. The main shaft is connected with the connecting rod and CS6-1 hand mechanism through the handle. The contact knife is composed of two grooved copper plates. It not only increases the heat dissipation area of the contact knife, which is beneficial to reducing temperature, but also improves the mechanical strength of the contact knife, which improves the dynamic stability of the switch.

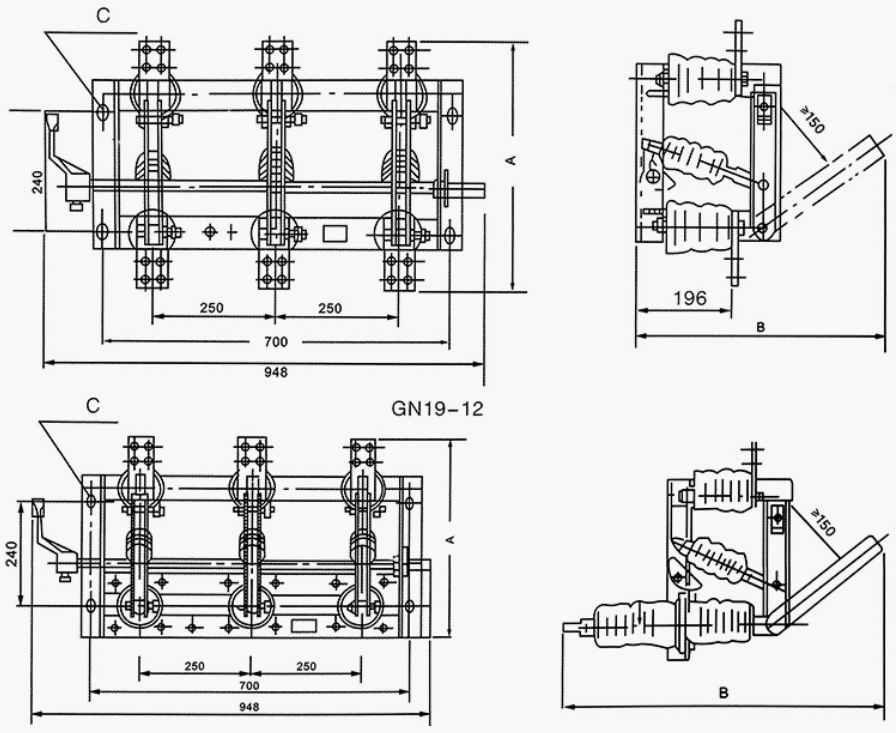
One end of the contact knife is installed on the contact base (or conductive rod) through bolts. In order to rotate the contact knife, the other end can be separately connected with the static contact knife system, and the contact pressure of the contact knife is maintained by the springs at both ends. A magnetic lock plate is arranged at the contact of the Richer indoor high-voltage disconnector. The purpose of adding the magnetic lock plate is to strengthen the attraction between the two slot contacts when a large short-circuit current passes through, that is, increase the contact pressure, thus improving the dynamic and thermal stability of the switch. The pole current of GN19-12/400A and GN19-12/630A disconnectors is small. Therefore, the structure does not need magnetic lock plate.

Base assembly: it is composed of base, main shaft, stopper, stop ring, etc. The limiter is mainly used to ensure that the required end position can be obtained when the conductive contact knife is opened and closed.

1.7 Outline and installation dimension



GN30-12(D) Disconnector installation dimension



GN19-12 Disconnector installation dimension

2.1 Overview of GN30-24(D) disconnectors

GN30-24 type rotary indoor high voltage disconnector is a new type of rotary contact switch. Its main structure is to fix two sets of insulators and contacts on the upper and lower two planes of the three-phase common base frame, through which the rotary contact is used. So as to realize the closing and opening of the switch.

GN30-24 switch is based on GN30-24 switch with earthing knife, which can meet the needs of different power systems. The product is compact in design, small in occupation space, strong in insulation capacity and easy to install and adjust. Its performance meets the requirements of GB1985-89 "AC high voltage disconnector and earthing switch". It is suitable for indoor power system with a rated voltage of 12kV/ AC50Hz and below, as the opening and closing circuit with or without voltage load. It can be used together with high-voltage switch cabinet or separately.

2.2 Product model description

G N 30 - 24 (D) / □ - □

- Thermal stability current (4s)
- Rated current
- With earthing device
- Rated voltage
- Design number
- Indoor
- Isolating switch

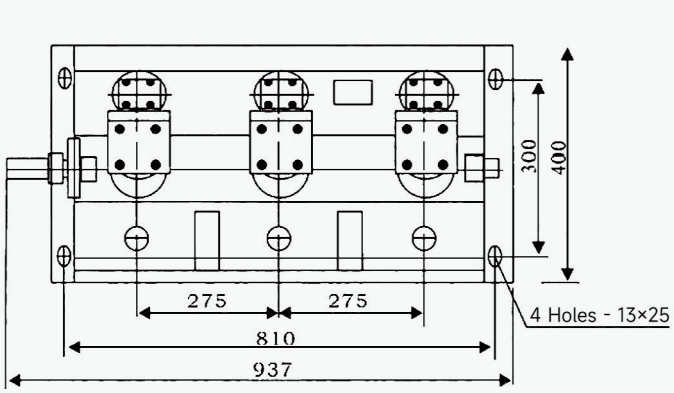
2.3 Use environment

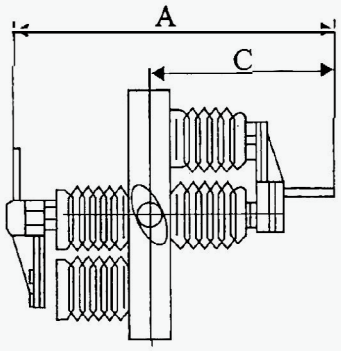
- Altitude not more than 1000 meters;
- Ambient temperature not more than+40℃ , no less than -10℃ ;
- The average relative humidity is not more than 95% per day and 90% per month;
- The seismic intensity shall not exceed 8 degrees;
- Places without serious dust, chemical corrosive and explosive substances;
- Places without frequent and violent vibration.

2.4 Technical parameter

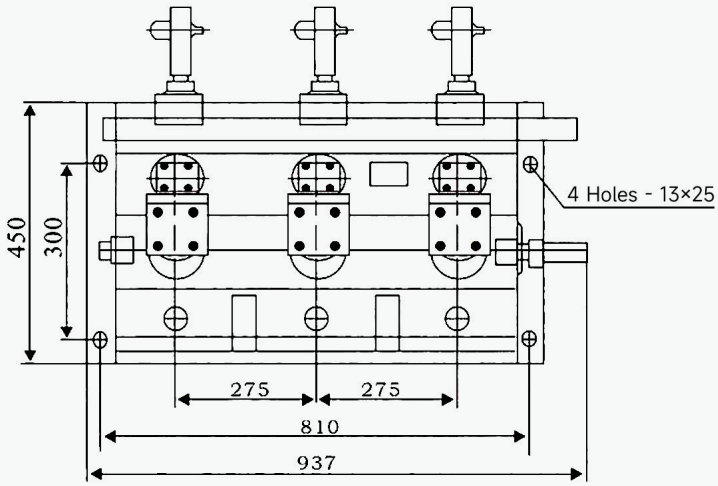
Item		Unit	Model and specification			
			GN30-24(D)/400-24.5	GN30-24(D)/630-20	GN30-24(D)/1000-31.5	GN30-24(D)/1250-31.5
Rated voltage		kV	12			
Maximum voltage		kV	12.5			
Rated current		A	400	630	1000	1250
Thermally stable current		kA	12.5	20	31.5	
Thermal stability time		S	4			
Dynamic stable current		kA	31.5	50	80	
Rated insulation level	Rated impulse voltage	kV	75			
	1min power frequency withstand voltage	kV	42			

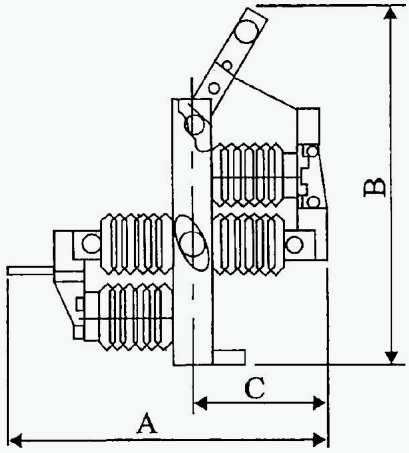
2.5 Outline and installation dimension



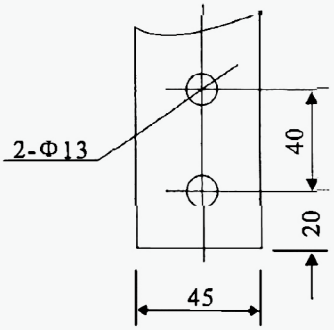


GN30-24/400(630, 1000) Outline and installation dimension

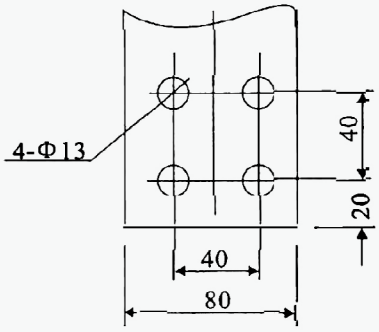




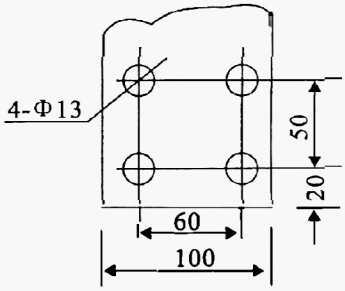
GN30-24D/400(630, 1000) Outline and installation dimension



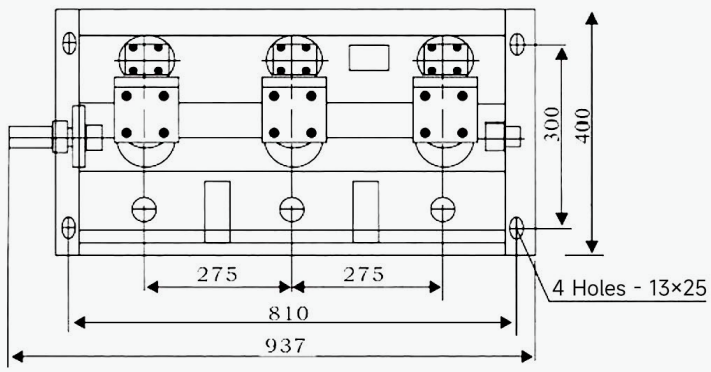
400A / 630A



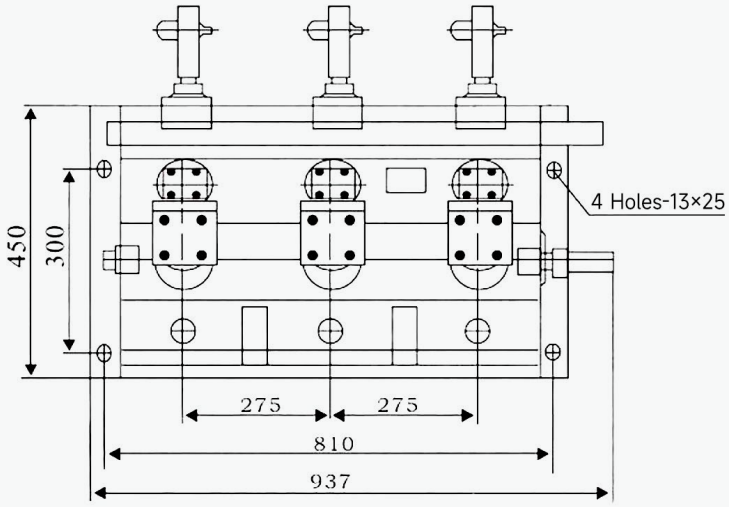
1000A



1250A



GN30-24/1250 Outline and installation dimension



GN30-24D/1250 Outline and installation dimension

GN30-24						GN30-24D								
400A / 630A		1000A		1250A		400A / 630A			1000A			1250A		
A	C	A	C	A	C	A	B	C	A	B	C	A	B	C
~560	~315	~580	~332	~580	~295	~550	~610	~240	~575	~615	~245	~580	~600	~295

03 Ordering instructions

- When ordering this switch, the technical data such as model, specification, rated voltage and rated current shall be indicated;
- If there are special requirements, they can be solved through consultation with the manufacturer.

E4.0701ME-202211

Zhejiang Greenpower Electric Co., Ltd
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Professional Power Equipment Provider

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